

MATERIALS SCIENCE

An anoxygenic photosynthesis-inspired dual biohybrid for light-enhanced chemical production

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Integrating light-harvesting materials with non-photosynthetic microbes offers a promising route to convert sunlight into chemicals. However, many existing biohybrid systems lack a clear architectural basis linking external light capture to intracellular redox processes. Inspired by the modular architecture of anoxygenic photosynthesis, we report a dual biohybrid system for light-enhanced chemical production in *Escherichia coli*. The system combines extracellular conjugated polymer nanoparticles as light-harvesting antenna with intracellularly biomineralized CdS nanoparticles as reaction-center-like charge-separation modules. This architecture supports light-enhanced malate production, yielding nearly a 30-fold enhancement relative to unmodified cells. Illumination-dependent studies reveal that activity is governed by light absorption, photon flux, and the intracellular redox environment. Fluorescence lifetime imaging microscopy further indicates photophysical coupling between the polymer and CdS components, consistent with energy transfer across the bio-synthetic interface and enhanced intracellular redox activity. Together, these results show how modular organization at the bio-synthetic interface can couple light harvesting with intracellular redox chemistry in living systems.

INTRODUCTION

Throughout billions of years of evolution, photosynthetic organisms, both oxygenic and anoxygenic, have developed sophisticated machinery to harvest sunlight and convert it into energy-rich chemical bonds (1). Their foundational architecture, in which light-harvesting antenna complexes funnel excitation energy into charge-separating reaction centers, has long inspired the design of artificial and semi-artificial platforms for solar-to-chemical conversion (1–4). Correspondingly, materials systems such as photoelectrochemical devices (5), photovoltaic-electrocatalyst assemblies (6), and semiconductor photocatalysts (2, 7) emulate this logic as artificial photosynthesis by absorbing sunlight, generating charge carriers, and channeling them toward thermodynamically uphill chemical reactions.

Alongside advances in artificial materials platforms, integrating light-absorbing materials with living cells offers unique opportunities for solar-to-chemical synthesis. Semi-artificial photosynthesis or biohybrid designs channel photogenerated charges directly into microbial intracellular redox networks, leveraging their metabolic versatility, catalytic selectivity, and adaptive capacity. By merging these biological strengths with the tunable photo-physics of synthetic semiconductors, such biohybrid systems expand the landscape of solar-driven chemical transformations beyond what either component can accomplish alone (8–12). Existing biohybrid photosynthetic platforms generally fall into two structural classes, defined by how the functional material interfaces with microbial cells. One approach integrates external light-harvesting materials, such as conjugated polymers (13, 14), cadmium telluride (15), metal sulfide (11, 16–18) or metal oxides nanoparticles (19, 20), or others (21), on the cell surface. While these materials provide strong and tunable optical absorption, their extracellular positioning often limits direct electron delivery into intracellular redox pathways. A second approach

embeds semiconductors within microbial cells through biomineralization (22) or intracellular uptake (23, 24), creating intimate contact between photoactive particles and native metabolic networks. This internal configuration can improve charge-transfer efficiency but remains constrained by limited control over particle formation, optical properties, or low uptake efficiency (25). Together, these strategies illustrate how synthetic light harvesters can activate microbial metabolism for chemical production. However, in many existing biohybrid systems, the relationship between external light harvesting and intracellular redox processes remains incompletely defined. In particular, how multiple photochemical components might be spatially organized and functionally coupled within a single biohybrid architecture has received limited experimental attention (26, 27). While previous studies have demonstrated semiconductor-microbe biohybrids, intracellular CdS biomineralization, and polymer-based light-harvesting systems individually, these components have typically been implemented in isolation. Addressing this gap may provide additional opportunities to examine potential interactions between distinct light-harvesting and charge-separation modules and their collective impact on light-enhanced chemical production.

Inspiration can be drawn from anoxygenic photosynthesis, one of nature's most ancient and modular solar-to-chemical architectures (28). Unlike oxygenic photosynthesis that operates two coupled photosystems and a Z-scheme dependent on water oxidation to sustain electron flow, anoxygenic photosynthesis relies on a single reaction center supported by peripheral antenna complexes (29). These antenna modules absorb light and funnel excitation energy directly into the reaction center, where charge separation occurs (30, 31), and electrons are further transferred into membrane-embedded quinone or Fe-S pathways to power downstream metabolism (32). This streamlined architecture, characterized by spatial separation of light harvesting and charge separation yet strong electronic coupling between the two, offers a conceptually tractable model for synthetic biohybrids (Fig. 1A). Guided by this principle, we engineered a synthetic analogue in which extracellular conjugated polymer nanoparticles mimic antenna complexes, while intracellularly biomineralized CdS nanoparticles function as reaction-center-like charge separators in non-photosynthetic microbes (Fig. 1A). The polymers, whose

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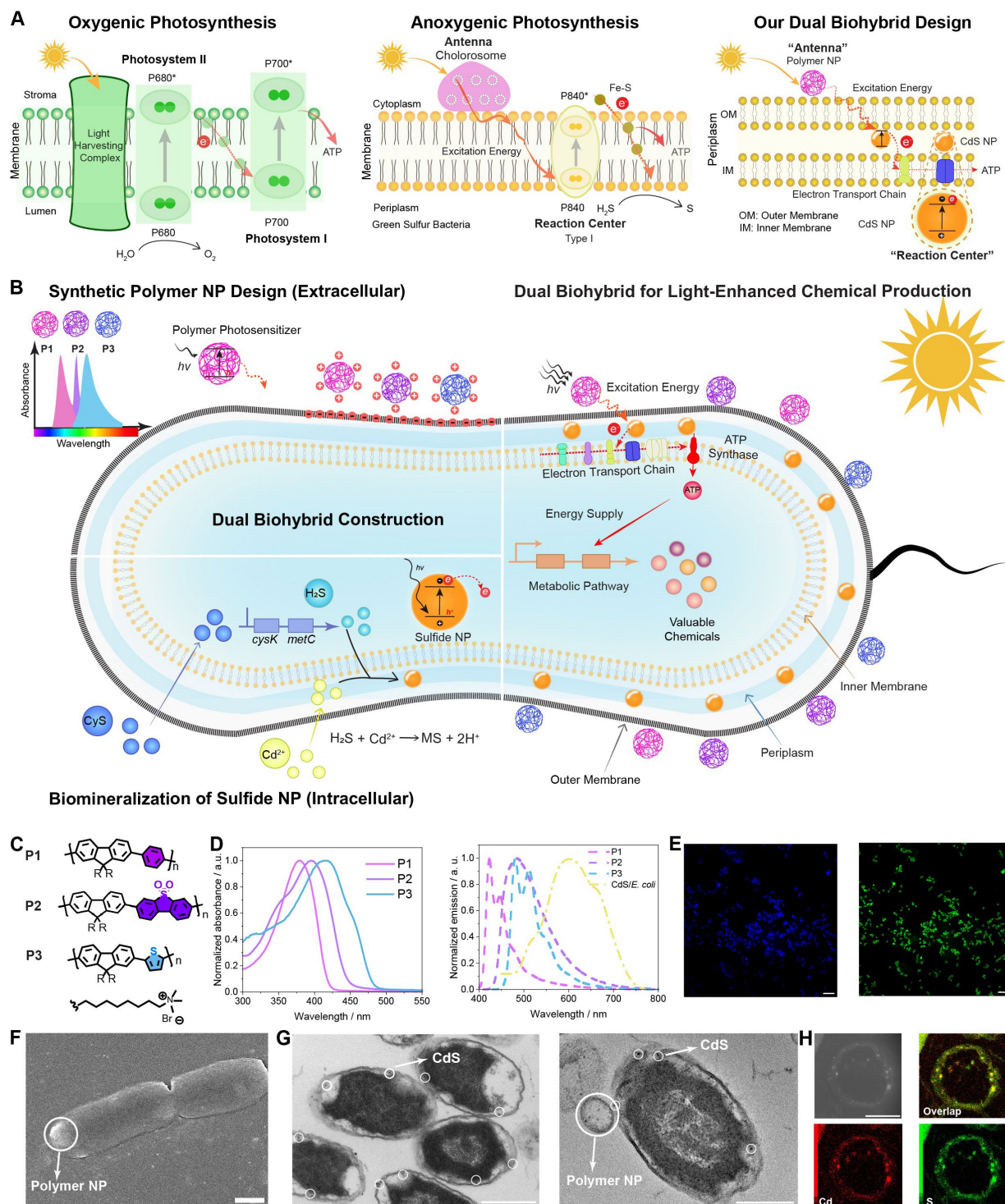


Fig. 1. Design of dual biohybrids mimicking anoxygenic photosynthesis for light-enhanced chemical production. (A) Schematic illustration showing the mechanism of oxygenic photosynthesis, anoxygenic photosynthesis and our dual biohybrid design. (B) Schematic illustrating dual biohybrid concept integrating extracellular synthetic conjugated polymer nanoparticles (P1–P3) with intracellularly biomineralized CdS nanoparticles for light-enhanced chemical production. Polymer nanoparticles act as extracellular light-harvesting antenna, transferring excitation energy to intracellular CdS (reaction centers), which interface with the bacterial electron transport chain to drive ATP generation and metabolic conversion. (C) Rational design of polymers, including P1, P2 and P3. (D) The normalized UV–Vis absorption spectra and emission spectra as well as their integration with *E. coli*. (E) Confocal fluorescence images of CdS/*E. coli* incubated with polymer nanoparticles. Scale bar 10 μm. (F to G) SEM and cross-sectional TEM images showing extracellular polymer nanoparticles (purple circles) and periplasmic CdS nanoparticles (orange circles) in the dual biohybrids. Scale bar, 500 nm. (H) High-resolution cross-sectional TEM and EDS elemental mapping confirming Cd and S colocalization within intracellular nanoparticles. Scale bars, 500 nm. Polymer nanoparticles were created in BioRender. Yang, Y. (2026) <https://BioRender.com/b6lja5r>.

absorption and emission properties can be systematically tuned, harvest incident photons and relay excitation energy to CdS across the cell envelope through FRET-like non-radiative energy transfer. In parallel, periplasm-localized CdS nanoparticles interface intimately with the bacterial electron-transport chain, enabling efficient charge separation and direct electron injection into native metabolic pathways (Fig. 1B).

Using the *E. coli* as a model chassis, we observed that this dual architecture markedly enhances light-enhanced chemical conversion, yielding nearly a 30-fold increase in malate production relative to cells alone. Systematic illumination studies revealed that performance is governed by the interplay between polymer absorption characteristics, photon flux, and intracellular redox energetics, highlighting both the design logic and the complementary functions of the polymer antenna and CdS reaction centers. Fluorescence lifetime imaging microscopy further revealed photophysical coupling between the extracellular polymer and intracellular CdS components, consistent with non-radiative polymer-to-CdS energy transfer analogous to excitation funneling in natural antenna systems. This synergistic energy–electron relay supports efficient light-enhanced malate production by coordinating photophysical interactions at the material–cell interface with intracellular redox processes. By integrating tunable synthetic antenna with biomineralized intracellular semiconductors, this work provides a modular experimental platform to examine how distinct photochemical components can be spatially organized and functionally coupled within non-photosynthetic microbes. More broadly, it illustrates how architectural principles inspired by anoxygenic photosynthesis, especially the separation and coupling of antenna and reaction-center functions, can inform the design of hybrid material–cell systems and provide mechanistic insights into light-enhanced modulation of microbial metabolism.

RESULTS

Construction of dual biohybrids with bio-interface design

To realize this dual biohybrid concept, we first synthesized a series of conjugated polymers with tailored optoelectronic properties and formulated them into nanoparticles for extracellular interfacing. In parallel, *E. coli* was processed to biomineralize CdS nanoparticles intracellularly, forming functional semiconductor domains within the periplasm. By assembling the polymer nanoparticles onto the bacterial surface, we constructed a dual biohybrid system that spatially integrates extracellular light-harvesting polymers with intracellular CdS reaction-center-like modules.

To systematically tailor light–matter interactions in biohybrid systems, we designed a series of structurally related conjugated polymers with varied optoelectronic properties. The central design objective was to modulate absorption behavior, through strategically incorporating comonomers with distinct electronic and structural features. Specifically, we synthesized three conjugated polymers *via* Pd(0)-catalyzed Suzuki–Miyaura polycondensation by copolymerizing bis(8-bromo-*n*-octyl) fluorene with phenylene (P1), dibenzo[*b,d*]thiophene sulfone (P2), and thiophene (P3) (Fig. 1C) (14). These monomers were chosen not only for their differing electronic characteristics, ranging from electron-rich (P3) to electron-deficient (P2), but also for their distinct structural features, including differences in planarity, conjugation length, and steric profile. Phenylene functions as a neutral, rigid spacer; dibenzo[*b,d*]thiophene sulfone introduces a strong electron-withdrawing group and a fused-ring geometry that

enhances backbone planarity; and thiophene provides electron-donating character and promotes π -conjugation. This framework allows us to systematically investigate how both electronic and structural parameters contribute to the modulation of optical gaps and the overall absorption profile of the copolymers. Because the bacterial surface is typically negatively charged (primarily from lipopolysaccharides, cell wall peptidoglycan, and phospholipids), the initially neutral conjugated polymers were functionalized with trimethylammonium groups to introduce positive surface charges, thereby enabling extracellular interfacing based primarily on electrostatic interactions (33, 34). Successful modification of the polymers was confirmed by ^1H NMR spectroscopy (supplementary information). Following functionalization, the polymers were processed into nanoparticles using a reprecipitation method (35), yielding uniform colloids with average size ranging from 107 to 124 nm (fig. S1, table S1). These dimensions are well-aligned with the bacterial scale, supporting effective extracellular integration at the material-microbe bio-interface. The resulting nanoparticles exhibited zeta potentials of up to +29 mV (table S1), confirming successful surface functionalization and enabling strong electrostatic association with negatively charged *E. coli* cells.

Backbone engineering produced polymer nanoparticles with distinct absorption profiles dictated by the incorporated comonomers. All nanoparticles displayed broad visible-light absorption, with P2 and P3 showing progressively red-shifted absorption onset compared to P1 (Fig. 1D). The introduction of dibenzo[*b,d*]thiophene sulfone (P2) leads to a red-shifted absorption profile, attributed to improved backbone planarity and weak donor-acceptor interactions with the fluorene unit. The thiophene-containing copolymer (P3) shows the most pronounced redshift and the narrowest optical gap. This is attributed to the thiophene's strong electron-donating nature, which raises the highest occupied molecular orbital (HOMO) level and its ability to enhance π -conjugation due to favorable planarity and minimal steric hindrance. This enhanced delocalization and conformational planarity also contribute to broader accessible electronic transitions, resulting in a widened absorption spectrum (36). Correspondingly, all polymers emitted across a wide spectral range (400–800 nm), providing a flexible platform for tailoring photoexcitation and energy transfer in microbial hybrid systems (Fig. 1D). To assess the intrinsic photoactivity of the polymers, we performed photocurrent measurements on drop-cast films under chopped white light irradiation (50 mW cm^{−2}, 30 s on/off cycles). All three polymers generated detectable anodic photocurrents compared to the dark baseline (fig. S2), confirming their ability to generate charge carriers upon excitation.

In parallel, to enable intracellular semiconductor formation, we leveraged the natural sulfur metabolic pathway of *E. coli* to direct the biosynthesis of CdS nanoparticles within the periplasmic space (*E. coli*/CdS). Upon supplementation with Cd²⁺, endogenous cysteine metabolism leads to H₂S generation, which reacts *in situ* with Cd²⁺ to nucleate CdS nanoparticles (22). Notably, biomineralization occurs predominantly in the periplasm, a confined oxidative compartment situated between the inner and outer membranes, rather than in the cytoplasm. This spatial localization is advantageous for several reasons: (i) it allows for proximity to the outer membrane, facilitating interaction with extracellular synthetic materials; (ii) it minimizes interference with cytoplasmic metabolic pathways; and (iii) most importantly, it situates the nanoparticles near the inner membrane, where the electron transport chain resides, thereby promoting efficient

electron injection to drive downstream energy conversion and bio-synthetic pathways.

Multiple imaging techniques confirmed the successful formation of dual biohybrid architecture. Confocal fluorescence imaging of the polymer/CdS/*E. coli* biohybrid reveals strong fluorescence signals associated with the bacterial cells (Fig. 1E). Control experiments show that both polymer/*E. coli* and CdS/*E. coli* samples produce fluorescence signals with similar spatial distributions (fig. S3), indicating that the observed fluorescence in the combined system arises from contributions of both polymer and CdS components. These results support the spatial co-localization of polymer nanoparticles with CdS-containing *E. coli* cells in the dual biohybrid. Scanning electron microscopy (SEM) further visualized polymer nanoparticles adhered to the bacterial surface (Fig. 1F), consistent with electrostatic assembly. Cross-sectional transmission electron microscopy (TEM) images revealed dense electron-contrast nanoparticles localized primarily in the periplasmic region, confirming the *in-situ* biomineralization of CdS nanoparticles (Fig. 1G) as well as the well-defined bio-interface and physical proximity between the polymer nanoparticles and CdS nanoparticles (Fig. 1G). High-resolution TEM coupled with energy-dispersive x-ray spectroscopy (EDS) mapping demonstrated the elemental identity of these nanoparticles, showing clear co-localization of cadmium (Cd) and sulfur (S) signals, thereby verifying the composition and spatial organization of the intracellular semiconductor component (Fig. 1H and fig. S4). In parallel, those CdS nanoparticles can be extracted and subjected to spectroscopic and structural analysis (note S1, fig. S5). The Ultraviolet–Visible (UV–Vis) absorption spectrum showed a sharp absorption edge at ~500 nm with strong absorption in the UV–blue region and a gradual tail into the visible range (fig. S5A). Together, these complementary imaging analyses confirm the successful construction of the dual biohybrid system, with extracellular polymer nanoparticles interfacing with the bacterial surface and intracellular CdS nanoparticles localized within the periplasm. This spatially organized architecture provides a well-defined platform to investigate cooperative light-harvesting and interfacial photochemical interactions across the bio-synthetic interface.

Dual biohybrids exhibit enhanced chemical production

After confirming the successful construction and spatial organization of the dual biohybrid system, we next evaluated its functional performance by examining light-enhanced biosynthesis of malate as a model output. Malate, a four-carbon dicarboxylic acid, is a valuable intermediate derived from biomass and widely utilized in fine chemical synthesis and as a food industry additive (37). In *E. coli*, malate can be produced through the reductive tricarboxylic acid cycle, a process that requires ATP input (38). Notably, elevated intracellular ATP levels can also activate key enzymes such as phosphoenolpyruvate synthase and pyruvate carboxylase, thereby redirecting carbon flux from pyruvate toward malate synthesis (38). This system provides a sensitive readout of intracellular energy availability and reductive capacity, making it a suitable platform to assess the bioenergetic enhancement conferred by the dual biohybrid under light stimulation.

We first evaluated the light-enhanced malate production of polymer/CdS/*E. coli* dual biohybrids under white light irradiation (peak at 450 nm, range 400–800 nm, intensity: 4.5 mW cm⁻²), and we confirmed that all polymer/CdS/*E. coli* dual biohybrid groups exhibited increased malate production compared to *E. coli* and CdS/*E. coli* control as well as their respective dark controls with up to 6.8-fold

increase (Fig. 2A). The corresponding polymer/*E. coli* controls are shown in fig. S6. These measurements together constitute a complete control matrix for evaluating the individual and combined contributions of polymer, CdS, and illumination. This marked improvement highlights the synergistic effect of the dual biohybrid configuration and underscores how polymer light-harvesting and polymer–CdS coupling collectively enhance electron transfer into the cell. To probe whether interfacial photophysical coupling influences intracellular bioenergetics, intracellular ATP levels and NADH/NAD⁺ ratios were measured under white-light illumination. Both metrics increased progressively from *E. coli* to CdS/*E. coli* and further to the polymer/CdS/*E. coli* dual biohybrid (fig. S7). In particular, the dual biohybrid exhibited the highest ATP levels and NADH/NAD⁺ ratio, indicating enhanced cellular energy status and reducing-equivalent availability. These results support the interpretation that polymer-assisted light harvesting, and subsequent CdS-mediated electron delivery alter intracellular bioenergetics, providing a biological basis for the enhanced malate production observed in the dual biohybrid system.

The light-enhanced metabolic activity in the dual biohybrid system arises from a coordinated interplay of photophysical and redox interactions between synthetic nanomaterials and microbial electron transfer pathways. To clarify this behavior, we outline key mechanistic routes that contribute to enhanced performance, including (1) polymer pathways, (2) CdS direct pathway, and (3) polymer–CdS coupled pathway. (i) To assess whether polymers alone could act as effective light harvesters (in the absence of intracellular CdS) for direct electron transfer, we constructed polymer/*E. coli* biohybrids and evaluated their malate production under identical conditions. However, malate production showed minimal enhancement (11.3–14.0 mg liter⁻¹ under irradiation vs 8.2–9.4 mg liter⁻¹ in the dark; fig. S6), indicating that direct polymer-to-cell electron transfer is inefficient under tested conditions. (ii) Biomineralized CdS nanoparticles, localized within the periplasmic space of *E. coli*, are strategically positioned adjacent to the inner membrane-bound electron transport chain. Upon illumination, CdS absorbs visible light and efficiently generates photo-excited electrons that can be transferred to the inner membrane electron transport chain, thereby enhancing cellular reducing power and boosting microbial biosynthetic output. (iii) In contrast, when light is absorbed by the extracellular conjugated polymer nanoparticles, multiple complementary pathways may be engaged (Fig. 2B). (i) In the presence of sacrificial electron donors (SEDs) such as cysteine or ascorbic acid, the photoexcited polymer may inject electrons directly into redox-active membrane components on the bacterial surface, thereby augmenting the intracellular electron pool. (ii) Alternatively, the electrons generated in the polymer may be relayed to adjacent CdS nanoparticles, which act as intermediate electron mediators owing to their physical proximity and favorable band alignment. This electron cascade may enhance charge separation and prolong carrier lifetimes, contributing to the improved photoactivity of the dual biohybrid. (iii) In addition to electron transfer, non-radiative energy transfer from the excited polymers to adjacent CdS may occur, enabled by their spectral overlap and close spatial proximity. This photosensitization of CdS may increase its excitation and strengthen photocatalytic electron injection, supporting improved chemical production. Together, these complementary processes, including direct and indirect electron injection, non-radiative energy transfer, and intracellular redox enhancement, indicate that multiple photophysical pathways contribute to the interaction between extracellular polymers and intracellular CdS nanoparticles. However, the relative contribution

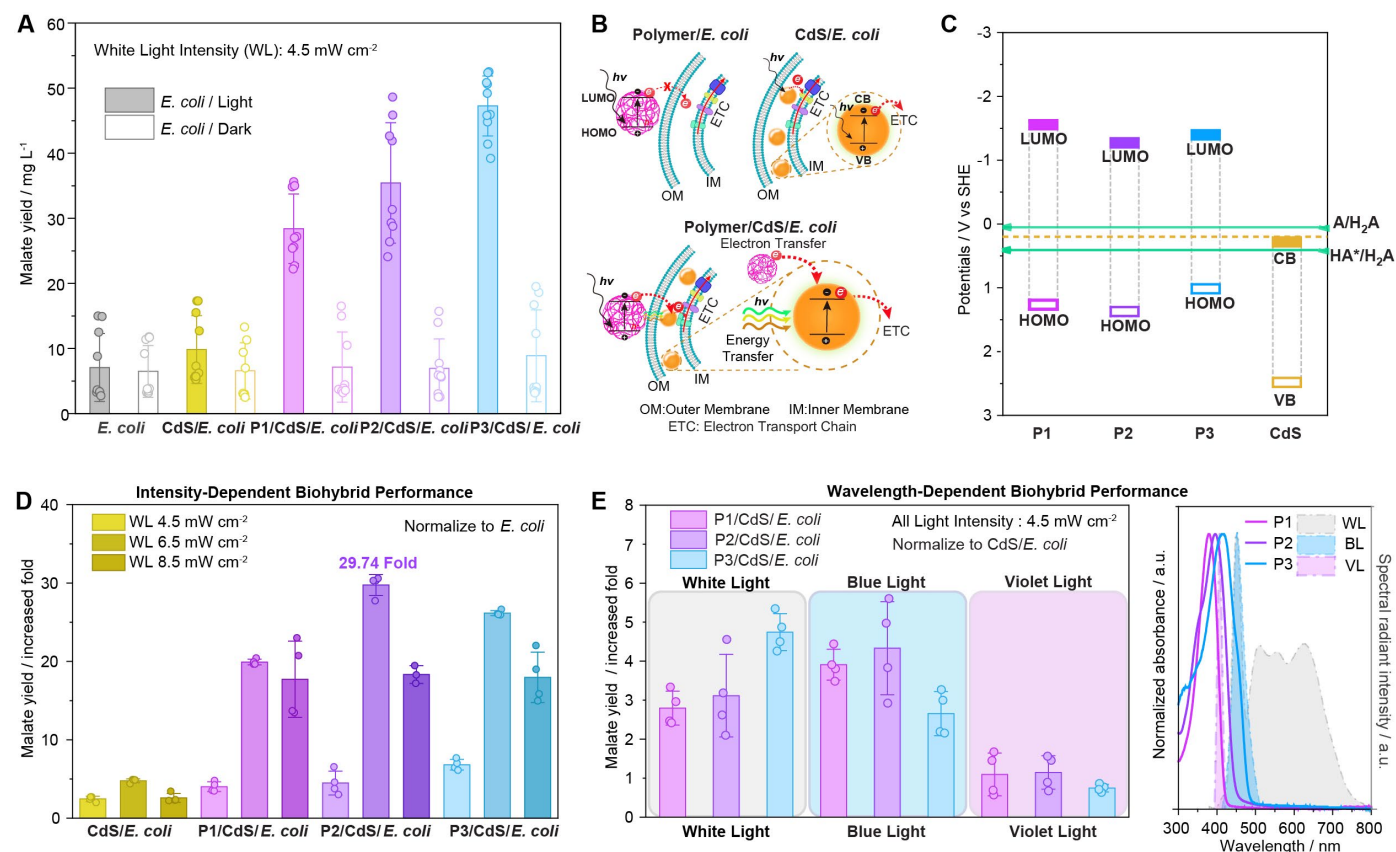


Fig. 2. Photoactivity and mechanistic basis of polymer/ $\text{CdS}/E. coli$ dual biohybrids. (A) Malate yield of $E. coli$, $\text{CdS}/E. coli$, and polymer (P1–P3)/ $\text{CdS}/E. coli$ biohybrids under white light irradiation (4.5 mW cm^{-2}) and dark conditions. The corresponding polymer/ $E. coli$ controls are shown in fig. S6. Together, these measurements constitute a complete control set for evaluating the individual and combined contributions of polymer, CdS, and illumination, and highlight the enhanced performance of the dual biohybrid system. (B) Schematic illustration of possible light-enhanced pathways: (top) individual polymer/ $E. coli$ or $\text{CdS}/E. coli$ systems, and (bottom) dual biohybrids where polymer excitation transfers energy or electrons to CdS, which subsequently injects charges into the bacterial electron transport chain (ETC). (C) Band alignment of polymers P1–P3 and CdS relative to the redox potentials of ascorbate/dehydroascorbate (A^-/HA , $\text{HA}^+/\text{H}_2\text{A}$) couples, indicating feasible electron transfer from polymer to CdS. (D) Light-intensity-dependent malate yields under white light at 4.5, 6.5, and 8.5 mW cm^{-2} , normalized to the $E. coli$ control. All three dual biohybrids show enhanced yields with increasing light intensity. The dual biohybrid can achieve an increase of almost 30-fold (29.74-fold) compared to $E. coli$ control. (E) Wavelength-dependent malate yields under white light (WL), blue light (BL, 455 nm), and violet light (VL, 405 nm) illumination, normalized to the $\text{CdS}/E. coli$ control. The right panel shows normalized polymer absorption spectra overlaid with lamp spectral profiles. Error bars denote standard deviations from at least four independent replicates. Polymer nanoparticles were created in BioRender. Yang, Y. (2026) <https://BioRender.com/b6lja5r>.

of each pathway to the overall light-enhanced malate production remains to be fully elucidated.

Although it is not practical to directly separate the relative contributions of electron and energy transfer in the photocatalytic assays, the energy level diagram (Fig. 2C, fig. S5, E to F, table S2) indicates that polymer-to-CdS electron injection is energetically feasible: the polymer LUMO levels (-1.3 V to -1.6 V vs. SHE) lie more negative than the CdS conduction band edge ($+0.2 \text{ V}$ vs. SHE), enabling downhill electron transfer into CdS. In contrast, the polymer HOMO levels ($+1.0 \text{ V}$ to $+1.4 \text{ V}$ vs. SHE) are far less positive than the CdS valence band ($+2.5 \text{ V}$ vs. SHE), rendering hole transfer energetically inaccessible. This energetic alignment provides a plausible mechanistic basis for enhanced metabolic activity in the dual biohybrid, arising from the synergy between extracellular polymers that harvest light and intracellular CdS nanoparticles that funnel electrons into microbial metabolic pathways.

To gain deeper insight into how polymer nanoparticles influence light-enhanced metabolic activity, we compared malate production

across the three dual biohybrids. Among the polymers, the P3/ $\text{CdS}/E. coli$ exhibited the highest malate yield ($47.6 \text{ mg liter}^{-1}$), corresponding to a $> 60\%$ increase over P1 ($28.4 \text{ mg liter}^{-1}$) and P2 ($35.4 \text{ mg liter}^{-1}$), and more than a 5-fold enhancement relative to $\text{CdS}/E. coli$ ($9.2 \text{ mg liter}^{-1}$) and over 7.4-fold relative to native $E. coli$ ($6.4 \text{ mg liter}^{-1}$) controls (Fig. 2A). These results demonstrate that polymer chemistry is a key determinant of dual biohybrid efficiency.

Polymer nanoparticles exhibit comparable size distributions (107–124 nm) that are broadly matched to bacterial dimensions. However, follow-up control experiments using smaller nanoparticles did not result in substantial improvement in malate production (fig. S8), suggesting that polymer nanoparticle size alone is unlikely to be the dominant factor governing performance within this system. Instead, the observed differences are likely governed more strongly by intrinsic photophysical properties of the polymers. Because each step in the polymer-mediated enhancement—light harvesting, exciton generation, charge separation, interfacial electron transfer, and potential photoinhibition—is intrinsically light-dependent, differences in

polymer performance are expected to arise from how effectively each polymer is excited under specific illumination conditions and how efficiently this excitation is coupled to downstream processes. This motivated us to investigate how illumination parameters regulate polymer contributions to light-enhanced activity in the dual biohybrid system.

Illumination-dependent polymer contributions to light-enhanced chemical production

To disentangle these effects, we examined two key components of the illumination environment: (i) the incident light intensity, which sets the magnitude of the effective photon flux, and (ii) the illumination spectrum, which determines spectral overlap with polymer absorption.

We first analyzed how photon flux—governed by light intensity—regulates exciton generation and subsequent charge or energy transfer to CdS. Notably, high-intensity irradiation is well-documented to suppress *E. coli* metabolism (39, 40), and our observations were consistent with this trend: native *E. coli* cultures exhibited a decline in both malate production (6.4 to 3.1 mg liter⁻¹) and final OD₆₀₀ values (2.8 to 1.0) as white light intensity continuously increased from 4.5 to 8.5 mW cm⁻² (fig. S9). In contrast, malate yields from all three dual biohybrids firstly increased with rising light intensity up to 6.5 mW cm⁻², beyond which a decline was observed at 8.5 mW cm⁻², suggesting a transition from light-limited to photo-inhibited conditions (Fig. 2D). To mitigate for batch-to-batch variability in cell density and baseline metabolic levels, malate yields were normalized to the corresponding *E. coli* controls acquired under identical illumination, yielding a polymer-CdS enhancement factor. The dual biohybrids achieve up to 30-fold enhancement in light-enhanced malate production compared to *E. coli* alone under 6.5 mW cm⁻² irradiation conditions, showing strong synergy between polymer and CdS in enhanced exciton generation and charge transfer. Furthermore, to decouple polymer contributions, malate yields were normalized to the corresponding CdS/*E. coli* controls (fig. S10). At low intensity (4.5 mW cm⁻²), P3/CdS/*E. coli* exhibited the highest enhancement (4.7-fold), consistent with efficient light-harvesting under photon-limited conditions. At intermediate flux (6.5 mW cm⁻²), P2 (6.2-fold) marginally outperformed P3 (5.5-fold), reflecting its larger driving force for donor oxidation and stronger performance under moderate irradiation. At the highest flux (8.5 mW cm⁻²), the relative enhancements of P1–P3 converged (~7.0–7.3 fold), suggesting the onset of a photoinhibition-limited plateau. This saturation is unlikely to arise from inadequate polymer excitation; rather, it is likely to reflect limits in CdS conduction-band utilization and downstream cellular redox turnover (41, 42), beyond which excess electrons or excitation energy are dissipated through recombination or diverted toward reactive oxygen species (ROS) formation.

While light intensity governs the total photon available for excitation, the spectrum distribution of the illumination source determines how effectively each polymer can harvest those photons. We therefore next evaluated how illumination wavelength influences polymer at a fixed light intensity of 4.5 mW cm⁻² (Fig. 2E). Similarly, malate yields were normalized to the corresponding CdS/*E. coli* controls acquired under identical illumination, yielding a polymer-specific enhancement factor. Meanwhile, inductively coupled plasma (ICP) analysis confirmed that CdS biomineralization remained consistent across all illumination conditions—both under light and in the dark—thereby providing a reliable baseline for comparing photocatalytic performance among different samples (fig. S11). Interestingly, although

the spectral overlap between polymer absorption and lamp output is highest under violet light (Fig. 2E), followed by blue and then white, the measured malate yields did not follow this trend but instead displayed distinct polymer-dependent behaviors. P1 (3.9-fold) and P2 (4.3-fold) exhibited superior performance under blue light, consistent with their stronger spectral overlap and favorable HOMO levels that provide a robust driving force for sacrificial donor oxidation (Fig. 2C). In contrast, P3 achieved the highest enhancement under white light (4.7-fold), where its broader absorption band could be more effectively utilized.

Under violet-light illumination, all three polymers contributed minimally. Based on additional intensity-dependent experiments (fig. S12), this behavior is consistent with efficient direct excitation of CdS under violet light (405 nm), which becomes more prominent at higher light intensity and reduces the relative polymer contribution. A general illumination-induced physiological burden under high-energy violet light may additionally influence the absolute activity across all groups. Extending the wavelength range, an additional near-UV excitation condition (370 nm) produced polymer-dependent activity trends that differ from those observed under violet light (405 nm) but are broadly similar to those under blue-light (455 nm) illumination (fig. S13). This behavior is consistent with the stronger absorption of P1 and P2 in the near-UV region, which favors excitation of the polymer components relative to CdS under these conditions.

Together, these results indicate that system behavior is governed not simply by wavelength, but by which component is preferentially excited and how the polymer properties influence subsequent energy transfer. Across different illumination conditions, distinct polymers exhibit optimal performance regimes: under white light, P3 performs best at low flux; under blue light and at intermediate flux, P2 becomes dominant; and P1, although generally less effective, exhibits modest improvement under blue excitation. These trends highlight that effective polymer design for dual biohybrids requires balancing spectral overlap, redox driving force, and photostability to optimize light-enhanced metabolic performance.

Polymer-to-CdS energy transfer visualized by fluorescence lifetime imaging microscopy

Building on the illumination- and polymer-dependent activity trends described above, we next sought to uncover the underlying mechanisms governing photoactivation in dual biohybrids. Interestingly, control experiments performed without added SEDs still showed light-enhanced enhancement (fig. S14), indicating that mechanisms beyond direct electron relay contribute to the system and that the observed activity cannot be solely attributed to SED-mediated chemistry (note S2).

While endogenous redox-active species in *E. coli* may partially sustain charge transfer, the persistence of photocatalytic activity under SED-free conditions suggests a contribution from non-radiative polymer-to-CdS energy transfer. Given the spectral overlap between the polymer emission (Fig. 1D) and the absorption of extracted CdS (fig. S5A), together with their nanoscale proximity within the dual biohybrid assembly, we hypothesize that the polymers act as light-harvesting antenna. By funneling excitation energy to the intracellular CdS nanoparticles, they amplify CdS photoexcitation and, in turn, enhance the overall activity of the dual biohybrid system. Nevertheless, activity-based measurements alone cannot disentangle electron transfer from energy transfer, motivating further spectroscopic and mechanistic analyses.

To probe the potential energy transfer processes, we employed fluorescence lifetime imaging microscopy (FLIM) to monitor polymer–CdS interactions within the dual biohybrids. For these measurements, equal concentrations of CdS/*E. coli* and *E. coli* were incubated with P3 polymer nanoparticles, selected for their superior light-harvesting under low-flux conditions that match the FLIM excitation setting. Given the negligible emission of CdS below 500 nm (Fig. 1D), we first focused FLIM measurements in the 458–500 nm range to minimize spectral interference and isolate polymer-associated dynamics. Lifetime decay curves were analyzed using a three-exponential reconvolution model with globally shared lifetime constants (τ_1 – τ_3) shared across the dataset, while component amplitudes (A_i) were determined on a per-pixel basis. The use of a three-component model was validated through phasor analysis, improvements in reduced χ^2 , and inspection of fit residuals. The shortest lifetime component (τ_1) was assigned to P3, based on spatial localization as well as its agreement with P3/*E. coli* controls (Fig. 3A). Despite its short value

(~0.10–0.17 ns), τ_1 was reliably extracted, as confirmed by the instrument response function (IRF) widths (~0.173 ns at 458–500 nm; fig. S15A) and established thresholds for FLIM deconvolution ($\geq 0.5 \times$ IRF FWHM). The robustness of the fitting was further supported by strong photon statistics and low residuals (36, 37).

Representative FLIM images revealed a clear shortening of the fast lifetime component (τ_1) in P3/CdS/*E. coli* system (0.1037 ns) relative to P3/*E. coli* (0.1163 ns), a trend consistently reproduced across three independent replicates (figs. S16 and S17). This observation was corroborated by the amplitude-weight average lifetimes (τ_{avg}), which decreased from 0.170 ns for P3/*E. coli* to 0.131 ns for P3/CdS/*E. coli* (Fig. 3B). The multi-exponential fits yielded low residual deviations and excellent agreement with experimental decay curves across replicates, confirming the robustness of the analysis (Fig. 3C, fig. S18). To further validate the lifetime shortening, we applied phasor analysis, a fit-free method that maps decay dynamics into Fourier coordinates (note S3) (43, 44). Phasor plots from four

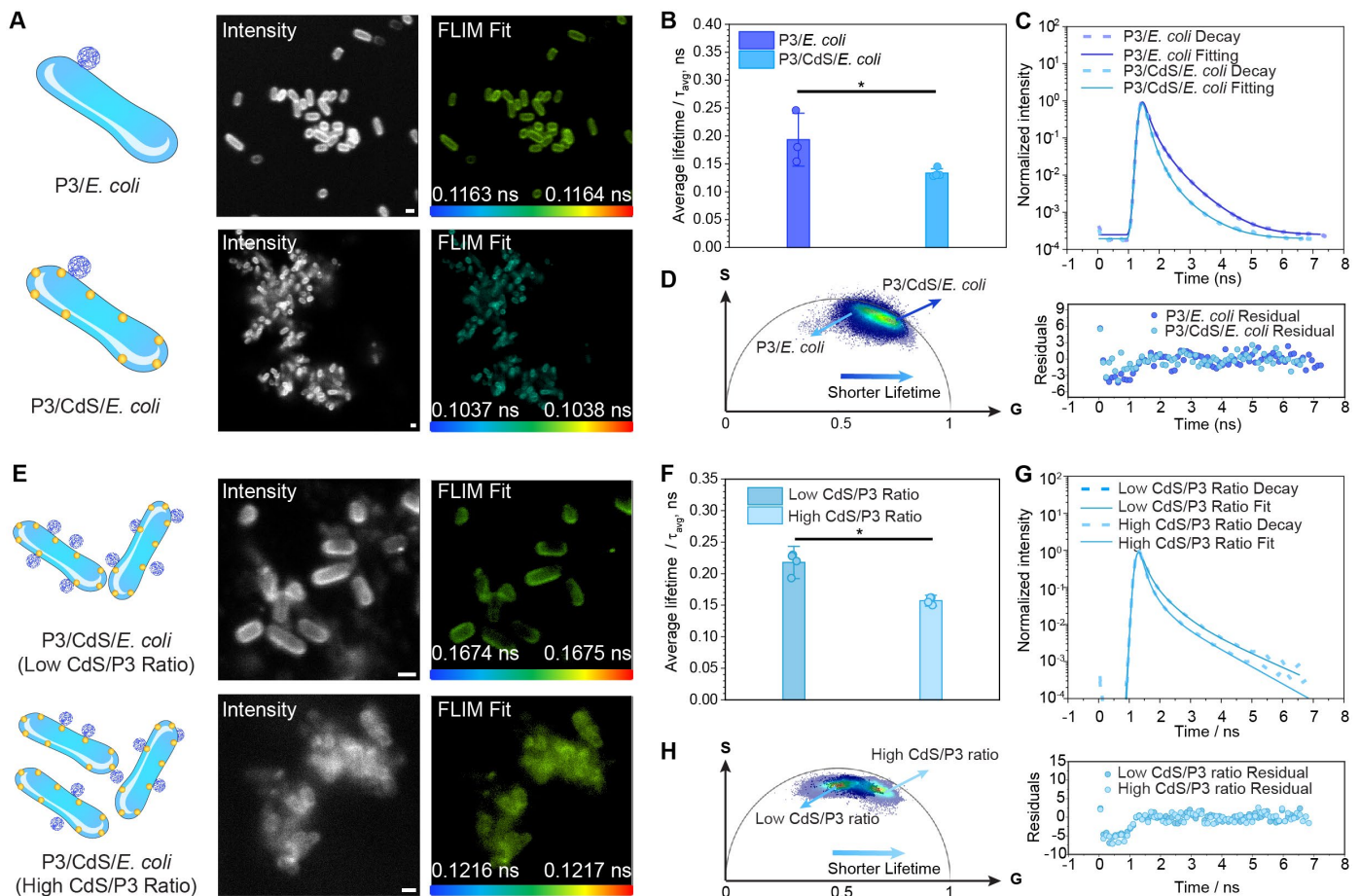


Fig. 3. Fluorescence lifetime analysis reveals polymer-to-CdS energy transfer in dual biohybrids. (A) FLIM intensity and fitted lifetime maps of *E. coli* associated with polymer P3 alone (top) or with dual P3/CdS hybrids (bottom). The lifetime of P3 decreased in the presence of CdS, indicating non-radiative coupling between the polymer donor and CdS acceptor. Scale bars, 1 μm . (B) Comparison of amplitude-weighted average lifetimes (τ_{avg}) for P3/*E. coli* and P3/CdS/*E. coli*. Data are mean $\pm$ S.D. from four replicates; * indicates $P < 0.05$. (C) Representative fluorescence decay curves (dashed line: decay curve; solid line: fitting and fitting residuals). (D) Phasor plots showing a shift toward shorter lifetimes for P3/CdS/*E. coli*, consistent with energy transfer to CdS. (E) FLIM images of P3/CdS/*E. coli* prepared with relatively low (top) and high (bottom) CdS/P3 ratios. The lifetime of P3 decreased with increasing CdS, indicating stronger non-radiative coupling between the polymer donor and CdS acceptor. (F) Corresponding τ_{avg} values showing stronger lifetime shortening at higher CdS/P3 ratio. (G) Fluorescence decay and fitting residuals for low vs high CdS/P3 ratios (dashed line: decay curve; solid line: fitting and fitting residuals). (H) Phasor plots confirming greater lifetime reduction in the high CdS/P3 ratio samples, indicative of stronger polymer-to-CdS energy transfer. Polymer nanoparticles were created in BioRender. Yang, Y. (2026) <https://BioRender.com/b6lja5r>.

replicates consistently show a leftward shift toward shorter lifetimes for the P3/CdS/*E. coli* samples (Fig. 3D), providing independent evidence of accelerated polymer relaxation dynamics. Additionally, control experiments verified that fluorescence contributions from *E. coli* and CdS were negligible (note S4, fig. S19), reinforcing that the lifetime shortening is attributable to interactions between polymer and CdS. In FLIM, such shortening signifies the emergence of an additional non-radiative decay pathway for the excited polymer, indicating the possibility of energy transfer from the extracellular polymer nanoparticles (donor) to the intracellular CdS nanoparticles (acceptor). In this context, the polymer serves as a light-harvesting antenna, while CdS functions as the energy acceptor that subsequently injects electrons into the bacterial redox network. Thus, the reduced polymer lifetime provides direct, quantitative evidence of polymer-to-CdS energy transfer, offering a mechanistic basis for the enhanced photocatalytic activity observed in the dual biohybrid systems.

To further elucidate polymer–CdS interactions, we extended the FLIM analysis to the full 458–800 nm emission range, where both polymer and CdS signals contribute, thereby better approximating the operational conditions of photocatalytic activity. Under these conditions, we maintained the polymer concentration (50 mg L^{-1}), while the amount of CdS/*E. coli* was varied to systematically increase the acceptor-to-donor (CdS-to-polymer) ratio. This design enabled us to examine how increasing CdS density and spatial proximity influence the polymer's excited-state dynamics, a characteristic feature of energy transfer. The instrument response function (IRF) for the 458–800 nm emission range had a FWHM of $\sim 0.219 \text{ ns}$ (fig. S15B). Global FLIM analyses first compared P3/CdS/*E. coli* with P3/*E. coli* at equivalent cell concentrations to assess whether CdS incorporation measurably altered polymer lifetimes under identical conditions. Within this detection window, CdS emission contributes to a longer-lived component that could partially offset polymer quenching (note S5, fig. S20). Nevertheless, the amplitude-weighted average lifetime (τ_{avg}) was shorter for P3/CdS/*E. coli* (0.157 ns) than for P3/*E. coli* (0.172 ns) (fig. S21), consistent with polymer quenching in the presence of CdS. Similar trends were consistently observed across replicates (figs. S22 and S23 for P3/*E. coli* and figs. S24 and S25 for P3/CdS/*E. coli*).

Having established this baseline effect, we next examined intra-group variations by systematically tuning the CdS loading. Increasing the CdS-to-polymer ratio led to a pronounced reduction in polymer lifetime, with τ_1 decreased from 0.1674 ns to 0.1216 ns (Fig. 3E). This progressive lifetime shortening strongly supports donor–acceptor coupling consistent with energy-transfer behavior. Correspondingly, amplitude-weighted average fluorescence lifetimes (τ_{avg}) decreased from 0.218 ns to 0.157 ns (Fig. 3F), while time-domain decay curves exhibited steeper slopes and excellent agreement with global reconvolution fits (Fig. 3G). Similar quenching trends were consistently reproduced across biological replicates at both high and low CdS-to-polymer ratio samples (figs. S24 to S27). Phasor plots from four replicates consistently showed a leftward shift under high CdS loading, confirming lifetime shortening (Fig. 3H), in agreement with enhanced polymer–CdS energy transfer driven by increased acceptor density. As a control, we also performed a benchmark experiment by varying the concentration of *E. coli* (no CdS) while keeping the P3 concentration constant. At high *E. coli* density, only a minor τ_{avg} decrease (from 0.202 to 0.172 ns, $\sim 15\%$) compared with low *E. coli* density (fig. S28). In contrast, increasing CdS/*E. coli* loading produced a

much larger τ_1 shortening ($\sim 27\%$), confirming that the observed lifetime reduction arises from specific polymer–CdS interactions rather than nonspecific cellular crowding effects (note S6). Collectively, these findings establish CdS as an effective energy acceptor that receives excitation energy from the polymer via non-radiative energy transfer. This process facilitates charge separation and subsequent electron injection into the microbial redox network, thereby enhancing the overall photocatalytic activity of the dual biohybrid system.

To summarize, in the polymer/CdS/*E. coli* dual biohybrid systems, polymer nanoparticles are associated with the bacterial cell surface, while CdS nanoparticles are intracellularly biomineralized and frequently located near the outer membrane, as indicated by confocal imaging and cross-sectional TEM. Considering the dimensions of the polymer nanoparticle (tens of nanometers) and the thickness of the bacterial membrane ($\sim 20\text{--}40 \text{ nm}$), the distance between polymer donors and intracellular CdS acceptors is expected to fall within the nanometers-to-tens-of-nanometer range. This spatial proximity is compatible with donor–acceptor coupling mechanisms, including FRET-like or radiative energy transfer pathways. Consistent with this interpretation, the observed shortening of the polymer fluorescence lifetime in the presence of CdS, and its dependence on CdS loading, supports the occurrence of interfacial energy transfer.

In this context, the relative loading of antenna-like donors and reaction-center-like CdS acceptors provides insights into the extent of coupling. Polymer nanoparticles were introduced at tens of mg liter^{-1} , whereas CdS was generated intracellularly from Cd^{2+} precursors ($\sim 3 \text{ mM}$), with ICP analysis indicating $\sim 2.7 \text{ mM}$ Cd associated with the cell pellet. Because CdS is confined within cells while polymer donors reside outside and interact with the cell surface, increasing CdS effectively increases the density of accessible acceptor sites, leading to progressively shorter polymer fluorescence lifetimes. In contrast, increasing polymer concentration produced little additional improvement (fig. S29), suggesting that donor–acceptor coupling is primarily limited by CdS availability rather than donor abundance. Although this architecture differs from natural anoxygenic photosynthetic systems—where antenna pigments typically outnumber reaction centers by orders of magnitude (45)—it captures the key principle of spatially separated donors and localized acceptors enabling interfacial energy transfer.

Dual excitation pathways governing polymer–CdS coupling

Motivated by the FLIM evidence for polymer-to-CdS energy transfer, we sought to consolidate how the two components of the dual biohybrid cooperate for light-enhanced chemical production. Prior analysis showed that polymers alone provide minimal enhancement (see Fig. 2B and fig. S6), CdS alone delivers moderate activity through direct electron injection, and the combined polymer/CdS configuration yields a markedly synergistic response. These observations indicate that polymer and CdS are functionally coupled: extracellular polymer harvest photons and either inject electrons directly or transfer excitation energy to intracellular CdS, which subsequently channels electrons into the bacterial redox network. This antenna–reaction-center-like organization sets up two overarching excitation routes—polymer-mediated excitation of CdS and direct photoexcitation of CdS—that operate to different extents depending on illumination conditions (Fig. 4).

The balance between these two routes depends on excitation wavelength, polymer energy levels, and the redox environments. White

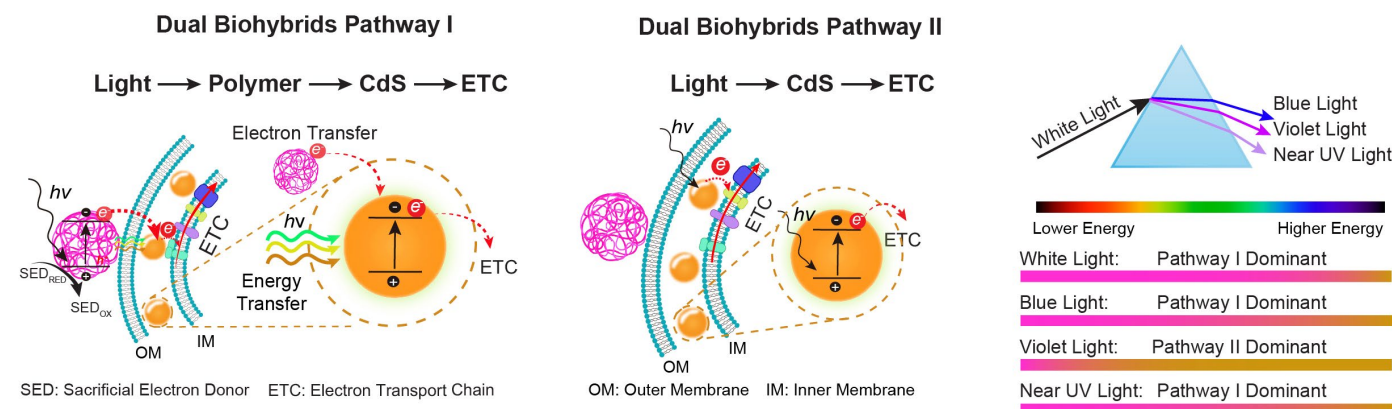


Fig. 4. Wavelength-dependent excitation pathways in polymer/CdS/*E. coli* dual biohybrids. Schematic illustration of two light-enhanced pathways contributing to intracellular electron delivery. Left (Pathway I): polymer-mediated route where photoexcited polymer nanoparticles transfer excitation energy or electrons to intracellular CdS, which subsequently injects electrons into the bacterial electron transport chain (ETC). Right (Pathway II): direct CdS photoexcitation and electron transfer to the ETC. Under white light, blue light (450 nm) and near-UV (370 nm) excitation, polymer-mediated contributions are favored, consistent with polymer absorption and coupling to CdS. In contrast, under violet light (405 nm), direct CdS excitation becomes more prominent, reducing the relative contribution of the polymer-mediated pathway. Polymer nanoparticles were created in BioRender. Yang, Y. (2026) <https://BioRender.com/b61ja5r>.

light and blue light (455 nm) align closely with the absorption tails of three polymers while minimizing direct CdS excitation. In contrast, the violet light (405 nm) more effectively excites CdS, consistent with the diminished polymer contribution under these conditions (Fig. 2D). Notably, an additional near-UV excitation condition (370 nm) produced polymer-dependent activity trends more comparable to those observed under blue-light illumination than under violet light (fig. S13). This behavior is consistent with enhanced absorption of P1 and P2 at shorter wavelengths, which favors polymer excitation relative to CdS.

Because the polymer and CdS are spatially separated by the cell membrane, long-range energy transfer is generally more feasible than direct electron tunneling, making polymer-mediated excitation of CdS favorable under white-, blue- and near-UV- illumination. Redox conditions further modulate these routes: lowering the SED concentration selectively suppresses polymer-to-CdS electron transfer by restricting polymer hole removal, whereas the energy transfer pathway remains partially operative (fig. S30). This trend mirrors the wavelength dependence, where decreased polymer excitation similarly reduces polymer-derived contributions to CdS excitation and malate yield.

Overall, these results indicate that the dual biohybrid operates as a spatially coordinated assembly in which extracellular conjugated polymers and intracellular CdS nanoparticles cooperatively expand the light-harvesting cross-section and promote charge transfer across the cell membrane. The combined contributions of non-radiative energy transfer and photoinduced electron transfer influence the efficiency with which absorbed light energy is converted into intracellular reducing power, with their relative roles modulated by excitation wavelength and redox environment. Together, this dual-pathway excitation behavior provides a mechanistic basis for the light-enhanced malate production observed in the polymer/CdS/*E. coli* biohybrid system.

DISCUSSION

This work demonstrates dual biohybrid architecture inspired by anoxygenic photosynthesis, in which spatially separated light-harvesting

and electron/energy-accepting components are functionally coupled to enable light-enhanced modulation of metabolism in a non-photosynthetic organism. By integrating extracellular conjugated polymer nanoparticles with intracellularly biomineralized CdS nanoparticles in *E. coli*, we create a cooperative interface that links photonic excitation to intracellular redox processes. In this system, the polymer acts as a light-harvesting donor that absorbs photons and becomes photoexcited, enabling photophysical coupling with intracellular CdS. CdS serves as the primary site of charge separation, generating photogenerated electrons that are subsequently delivered to intracellular redox pathways to support ATP and NADH production and downstream metabolism.

Systematic illumination studies, together with fluorescence lifetime imaging microscopy, indicate that enhanced activity is associated with photophysical coupling between the polymer and CdS components, consistent with non-radiative polymer-to-CdS energy transfer. This coupling enhances light utilization within the biohybrid system, resulting in up to a sevenfold increase in malate production compared with CdS/*E. coli* control. Given the spatial separation between extracellular polymer particles and intracellular CdS nanoparticles, direct electron tunneling between the two components is unlikely. The observed FLIM response is therefore interpreted as evidence of photophysical coupling consistent with polymer-to-CdS energy transfer rather than direct interfacial electron transport.

Taken together, these results support a working model in which polymer-mediated excitation enhances CdS photoactivity, and the resulting photogenerated electrons are coupled to intracellular redox metabolism, thereby increasing ATP/reducing-equivalent availability and promoting malate production. While the present data do not resolve every individual intracellular electron-transfer step, they establish mechanistic evidence between interfacial donor-acceptor coupling with enhanced metabolic output.

Although inspired by the functional organization of anoxygenic photosynthesis, the present biohybrid remains fundamentally heterotrophic. The demonstrated process is the light-enhanced conversion of glucose into malate, with glucose serving as both the carbon source and the primary metabolic energy source, while illumination

provides an auxiliary input that enhances intracellular photo-redox activity and redirects metabolic flux. Accordingly, the increased malate production under illumination should be interpreted as enhanced heterotrophic glucose conversion. The significance of this work lies in establishing the fundamental photophysical coupling between extracellular light-harvesting polymers and intracellular CdS nanoclusters and demonstrating that this coupling can enhance heterotrophic microbial production. This mechanistic understanding provides a basis for the future design and optimization of light-assisted biohybrid platforms with improved product yield and productivity under low-intensity illumination. However, quantifying the respective energetic contributions of glucose metabolism and illumination to malate production will require a complete carbon and energy balance, including measurements of absorbed photon flux, photochemical efficiency, glucose consumption, biomass and by-product formation, and intracellular metabolic fluxes. Similarly, a meaningful techno-economic analysis could potentially assess whether improvements in titer, yield, and productivity offset the costs associated with light delivery, biohybrid preparation, bioreactor operation, and downstream processing; such an analysis will be more appropriate after process-scale optimization and validation as future work.

More broadly, this work illustrates how principles inspired by anoxygenic photosynthesis, particularly the coupling of antenna and reaction-center functions, can guide the design of hybrid material-cell systems for light-enhanced metabolic modulation. We therefore view this platform as a mechanistic model from which more biocompatible, stable, and scalable systems may be developed in future work.

MATERIALS AND METHODS

General methods

All reagents were obtained from Sigma-Aldrich, Fluorochem, or TCI and used as received. Polymer P1, P2 and P3 were synthesized according to a previously published procedure (14) and fabricated into nanoparticles through the reprecipitation method (35). UV-Vis absorption spectra of the polymer nanoparticle solution were recorded on a Shimadzu UV-2550 UV-Vis spectrometer. The photoluminescence spectra of the polymer nanoparticle solution were measured with Edinburgh Instruments FLS1000. Dynamic light scattering and zeta-potential measurements were performed on a Malvern Zetasizer Nano Particle Sizer at 25°C. The emission spectrum of CdS/*E. coli* was obtained on a Leica STELLARIS 8 FALCON upright microscope with a 100×/1.4 oil objective.

Polymer synthesis

Polymers P1, P2, and P3 (without trimethylammonium-substitution) were synthesized by Suzuki coupling polymerization of the corresponding monomers following established procedures. In a typical reaction, the monomers, toluene, Starks' catalyst, and an aqueous solution of Na₂CO₃ were combined, degassed under N₂, and polymerized with [Pd(PPh₃)₄] as catalyst. Crude products were purified by Soxhlet extraction with methanol, acetone, and ethyl acetate. The high-molecular-weight fractions were recovered in chloroform, reprecipitated from methanol, and dried under reduced pressure.

For the synthesis of trimethylammonium-substituted polymers, a 0.1 g amount of the polymers above were dissolved in CHCl₃ (20 ml), and then a solution of trimethylamine in ethanol (30 wt %, 10 ml) was added. The reaction mixture was stirred at room

temperature for 48 h. The solvent was then evaporated, and the product was dried under vacuum to obtain the quaternized polymer.

P1: yellow solid in 78% yield. ¹H NMR (400 MHz, DMSO-d₆, δ): 8.65–7.80 (m, 10H, Ar-H), 3.35 (s, 4H, -CH₂-), 3.16 (br, 4H, -CH₂-), 2.98 (m, 18H, -CH₃), 1.53 (s, 4H, -CH₂-), 1.38–0.35 (m, 20H, -CH₂-). P2: light-yellow solid, 71% yield. ¹H NMR (400 MHz, DMSO-d₆, δ): 8.63–7.82 (m, 12H, Ar-H), 3.35 (s, 4H, -CH₂-), 3.18 (br, 4H, -CH₂-), 2.97 (m, 18H, -CH₃), 1.53 (s, 4H, -CH₂-), 1.37–0.35 (m, 20H, -CH₂-). P3: orange solid in 45% yield. ¹H NMR (400 MHz, DMSO-d₆, δ): 8.64–7.79 (m, 8H, Ar-H), 3.34 (s, 4H, -CH₂-), 3.17 (br, 4H, -CH₂-), 2.96 (m, 18H, -CH₃), 1.52 (s, 4H, -CH₂-), 1.38–0.34 (m, 20H, -CH₂-).

Polymer nanoparticle preparation

Conjugated polymers (P1, P2, and P3; 10 mg) were self-assembled into nanoparticles in water using a reprecipitation method. Briefly, each polymer was dissolved in THF (12.5 ml) under stirring, and 3 ml of the resulting solution was rapidly injected into 12 ml of deionized water under sonication at 30°C for ~10 s. The THF was removed by partial evaporation at 65°C for 5 hours, and the dispersion was filtered through a 0.45 μm nylon syringe filter (35).

Assembly of polymer/CdS/*E. coli* biohybrid systems

E. coli BL21(DE3) was cultured overnight in LB medium at 37°C. Subsequently, 1% of the bacterial culture was transferred into 10 ml fresh LB medium and grown aerobically at 37°C for ~3 hours to an OD₆₀₀ of 0.6–0.8. The cells were collected and resuspended in 10 ml M9 medium (pH 7.2) containing Na₂HPO₄·2H₂O (8.50 g liter⁻¹), KH₂PO₄ (3.0 g liter⁻¹), NH₄Cl (1.0 g liter⁻¹), NaCl (0.5 g liter⁻¹), CaCl₂ (0.1 mM), MgSO₄ (1.0 mM), and glucose (3.0 g liter⁻¹).

For CdS/*E. coli* groups, 1.0 mM cysteine and 0.3 mM CdCl₂ were added to induce intracellular CdS biomineralization. For polymer/CdS/*E. coli* groups, an additional 20.0 mg liter⁻¹ conjugated polymer nanoparticle solution and 1.0 mM ascorbic acid were introduced during light-enhanced malate production. All cultures were incubated aerobically at 37°C for 24 hours under shaking conditions (200 rpm), either under illumination or in the dark, depending on the experimental design.

After cultivation, the supernatants were collected for determination of Cd removal efficiency by ICP-OES (Thermo Scientific, iCAP 6000). The cell pellets were collected for subsequent characterization of the biohybrid systems and intracellular CdS nanoparticles.

SEM

The collected CdS/*E. coli* pellets were resuspended in 10 mM Tris-HCl buffer (pH 7.2) and mixed with 20 mg l⁻¹ P2 polymer nanoparticle solution. To prepare a specimen for SEM characterization, 10 μl of the mixture was pipetted on top of a silica disc mounted on a metal stub by conductive tape. Then, the dried specimen was imaged using SEM (JEOL, JSM-7610FPlus) after platinum spraying.

CdS nanoparticle extraction

30 ml *E. coli* was cultured with CdCl₂ and cysteine, and the cell pellets were collected after 24 hours, and then resuspended in 15 ml deionized water. The mixture was sonicated in an ice bath for 1 hour using a sonicator (UC-120) to achieve cell lysis. The resulting particles were separated by centrifugation at 13,000 rpm for 1 hour. To further modify the surface of the biosynthesized nanoparticles, the particles were dispersed in 5 ml Li₂S/formamide solution (5 g l⁻¹) by sonication for 15 min. Next, 10 ml didodecyldimethylammonium bromide

(DDAB)/toluene (50 g l⁻¹) was added to the top of the obtained uniform solution, followed by continuous stirring overnight. The DDAB-modified nanoparticles were collected from the oil layer, washed three times with acetone and ethanol to remove excess DDAB, centrifuged at 13,000 rpm for 10 min, and redispersed in toluene. The nanoparticles were characterized by transmission electron microscopy (TEM) (JEOL, JEM-2100) and x-ray powder diffraction (Rigaku, MiniFlex600).

TEM

For P2/CdS/*E. coli* cross-sectional sample, cell pellet was collected and fixed using 2.5% glutaraldehyde and placed at 4°C for 12–24 hours. The sample was then fixed with 1% OsO₄ solution for 1–2 hours. After removing OsO₄, the sample was rinsed in 0.1 M PBS (pH 7.4) 3 times, 15 min each, followed by dehydration, resin penetration and embedding, and polymerization. The resin blocks were trimmed to 70–90 nm thickness using an ultramicrotome (Leica, UC7), and the sections were transferred onto copper grids. The sample was stained with uranyl acetate for 8–15 min, followed by lead citrate staining for 5–10 min. After drying, the sample was imaged under TEM (HITACHI, HT7800) with 80 kV voltage and JEOL JEM-F200 with 200 kV acceleration voltage, respectively. EDS mapping was collected with JEOL JEM-F200.

Fluorescence imaging

Following the biohybrid assembly procedures, after 24-hour incubation with CdCl₂ and cysteine, the cell pellets were collected and resuspended in M9 medium with or without polymer nanoparticle solution. A drop of (~5 µl) biohybrid suspension was applied to a microscope slide, covered with a glass coverslip and sealed with nail polish. The fluorescence imaging was conducted on a Nikon Eclipse Ti inverted microscope with a 100×/1.45 plan apo DIC oil objective, and excitation wavelengths at 405 nm (emission bandpass filter, midpoint/bandwidth: 452/45) and 488 nm (emission bandpass filter: 571/72) were applied in imaging, respectively. Images were processed with the FIJI image processing package.

Photocurrent measurements

Photocurrent measurements were carried out in a one-compartment, three-electrode quartz cell using a 300 W xenon lamp (55 mW cm⁻², white light) as the light source. Illumination was controlled by a home-built timer to alternate 30 s on/off cycles in synchrony with the potentiostat (CHI-1000C). The electrolyte was 1 × PBS (pH 7.4) at 25°C. Fluorine-doped tin oxide (FTO, 1 × 1 cm) served as the working electrode; samples were prepared by drop-casting 100 µL of a 1 mg ml⁻¹ polymer suspension in ethanol or DMSO with 10% Nafion, drying under an IR lamp. A 1 × 1 cm Pt foil and an Ag/AgCl (saturated KCl) electrode were used as counter and reference electrodes, respectively. Photocurrents were recorded at 0.4 V vs. Ag/AgCl, and dark current was subtracted to yield the photocurrent density.

Semiconductor energy level determination

The ionization potentials (IPs) of the polymers determined by photoelectron spectroscopy in air (PESA) were taken from previous work (14), which directly yields the HOMO (highest occupied molecular orbital) energy relative to vacuum. The valence band maxima (VBM) of the inorganic semiconductors were obtained from ultraviolet photoelectron spectroscopy (UPS). VBM of CdS was obtained by UPS as the sum of the work-function-derived distance to vacuum

and the valence-edge onset relative to Fermi level (E_F). Optical gap (E_{opt}) was extracted from UV-Vis absorption onset. Because both PESA and UPS reference energies to the vacuum level, the measured HOMO and VB positions can be directly compared. The corresponding LUMO (lowest unoccupied molecular orbital) and conduction band (CB) positions were estimated by subtracting the optical gaps.

Light-enhanced malate production by biohybrid systems

E. coli BL21(DE3) was cultured overnight in LB medium at 37°C. Then, 1% bacteria culture was transferred into 10 ml fresh LB medium and grown aerobically at 37°C until an OD₆₀₀ of 0.6–0.8 was reached. Cells were collected and resuspended in 10 mL M9 medium (pH 7.2).

In a typical experiment, 1.0 mM cysteine and 0.3 mM CdCl₂ were added for CdS/*E. coli* group and an additional 20.0 mg l⁻¹ conjugated polymer (P1, P2, and P3) nanoparticle solution and 1.0 mM ascorbic acid were added for polymer/CdS/*E. coli* group. The *E. coli* cells were incubated aerobically at 37°C with shaking (200 rpm) under light irradiation (light intensity, 4–5 mW cm⁻²) or in the dark.

After 24 hours of cultivation, the supernatant was filtered and analyzed using high-performance liquid chromatography (HPLC) (Shimadzu, UFLC). Malate was quantified using an Aminex HPLX-87H column (300 mm by 7.8 mm, Bio-Rad) with detection at 210 nm. The mobile phase consisted of 5 mM H₂SO₄ at a flow rate of 0.6 ml min⁻¹.

Spectral characterization of CdS/*E. coli*

Emission spectra of CdS/*E. coli* were measured using Leica Stellaris 8 confocal microscope equipped with a tunable white light laser (WLL) and spectral detector. Emission spectra were recorded from 450 to 770 nm upon excitation at 440 nm. Spectral data were processed using Leica LAS X software.

Fluorescence lifetime imaging and analysis

Fluorescence lifetime imaging (FLIM) was performed using a Leica Stellaris 8 system equipped with a hybrid detector and pulsed white-light laser. *E. coli* cells were collected after 24 hours of incubation with/without CdCl₂ and cysteine, washed with an M9 medium twice to remove residual growth medium and added chemicals, and resuspended in an M9 medium with or without a polymer (P3) nanoparticle solution to maintain the same fluorescence background and pH values. CdS and P3 FLIM signals were collected using a 448 nm excitation (Supercom white light laser line with a notch filter) and 458–500 nm and 458–800 nm emission, respectively. High cell density (OD₆₀₀ > 5) and polymer concentration (50–100 mg l⁻¹) were used to collect enough photons for statistics. Excitation intensity and number of frames collected were adjusted to collect >100 photons/pixel while maintaining 1 photon per pulse.

Instrument response functions (IRFs) were recorded for each experimental batch using Leica LAS X software and used for deconvolution fitting in both image-based and time-domain fluorescence lifetime analysis. The IRFs were normalized and aligned to time zero based on the peak maximum. For experiments using the 458–500 nm emission range, the IRF full width at half maximum (FWHM) was measured to be approximately 0.173 ns, while for experiments using the broader 458–800 nm emission range, the IRF FWHM was ~0.219 ns. This consistency in temporal resolution across experiments ensured reliable separation of short fluorescence lifetimes.

FLIM data were analyzed using Leica LAS X software. A global three-exponential (three-component) deconvolution model was applied

to FLIM image stacks, incorporating the IRF to account for system temporal resolution and accurately extract τ_1 , τ_2 , and τ_3 for each pixel. Goodness-of-fit was evaluated using the reduced chi-square (χ^2) and residual analysis (see table S2 for fitted parameters and χ^2 values). Phasor plot analysis was performed using Leica LAS X software, with the harmonic frequency set to 9 to optimize the resolution of short lifetime components in the nanosecond range. Each pixel's fluorescence decay was transformed via Fourier analysis into G and S coordinates, generating a 2D phasor plot distribution (43). Pixel density was visualized using a heatmap scale (red: high, blue: low). To suppress background noise, a wavelet filter and intensity threshold were applied.

Intracellular ATP and NADH/NAD⁺ measurements

Intracellular ATP levels and NADH/NAD⁺ ratios were quantified under white-light illumination conditions using *E. coli*, CdS/*E. coli*, and polymer/CdS/*E. coli* samples prepared as described above. Cells were cultured to OD₆₀₀ of 0.6–0.8 and subjected to illumination under identical conditions as used for malate production experiments. Following illumination for 24 hours, cells were harvested by centrifugation and washed with ice-cold phosphate-buffered saline (PBS). ATP levels were measured using a luciferase-based ATP assay kit (Promega) according to the manufacturer's instructions. For NADH/NAD⁺ measurements, intracellular cofactors were extracted and quantified using a commercial NADH/NAD⁺ assay kit (Beyotime), following the manufacturer's protocol. Intracellular protein concentrations of the collected bacterial samples were determined using a Thermo Fisher Scientific BCA protein assay kit, and all ATP values were normalized to total protein content.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S30
Tables S1 to S3

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